

6601001123_Lab8

January 17, 2026

```
[21]: from keras.datasets import mnist
      from keras.utils import to_categorical
      from keras import layers, models, regularizers
      import numpy as np
      import matplotlib.pyplot as plt
```

```
[22]: (train_x, train_y), (test_x, test_y) = mnist.load_data()
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```
[23]: print("train_x.shape : ", train_x.shape)
      print("test_x.shape : ", test_x.shape)
```

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train_x.shape : (60000, 28, 28)
test_x.shape : (10000, 28, 28)
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```
[24]: train_x = train_x.reshape((train_x.shape[0], 28*28)).astype('float32') / 255
      test_x = test_x.reshape((10000, 28*28)).astype('float32') / 255
```

```
# train_x = train_x.astype('float32') / 255
# test_x = test_x.astype('float32') / 255
```

```
train_y = to_categorical(train_y)
test_y = to_categorical(test_y)
```

```
[25]: validation_x = train_x[50000:60000]
      validation_y = train_y[50000:60000]

      train_x = train_x[:50000]
      train_y = train_y[:50000]

      print("train_x.shape : ", train_x.shape)
      print("validation_x.shape : ", validation_x.shape)
      print("test_x.shape : ", test_x.shape)
```

```
train_x.shape : (50000, 784)
validation_x.shape : (10000, 784)
test_x.shape : (10000, 784)
```

```
[26]: def buildModel(neuronNum, train_x) :
      model = models.Sequential([
```

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        layers.Dense(neuronNum , activation='relu' , input_shape=(train_x.
↪shape[1],)) ,
        layers.Dense(neuronNum , activation='relu') ,
        layers.Dense(10 , activation='softmax')
    ])
    model.compile(optimizer='rmsprop', loss='categorical_crossentropy', ↪
↪metrics=['accuracy'])
    return model

```

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[27]: def buildModelRegularize(neuronNum,train_x) :
        model = models.Sequential([
            layers.Dense(neuronNum , kernel_regularizer=regularizers.l2(0.001) , ↪
↪activation='relu' , input_shape=(train_x.shape[1],)) ,
            layers.Dense(neuronNum , kernel_regularizer=regularizers.l2(0.001) , ↪
↪activation='relu') ,
            layers.Dense(10 , activation='softmax')
        ])
        model.compile(optimizer='rmsprop', loss='categorical_crossentropy', ↪
↪metrics=['accuracy'])
        return model

```

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[28]: def buildModelDropout(neuronNum,train_x) :
        model = models.Sequential([
            layers.Dense(neuronNum , activation='relu' , input_shape=(train_x.
↪shape[1],)) ,
            layers.Dropout(0.5) ,
            layers.Dense(neuronNum , activation='relu') ,
            layers.Dropout(0.5) ,
            layers.Dense(10 , activation='softmax')
        ])
        model.compile(optimizer='rmsprop', loss='categorical_crossentropy', ↪
↪metrics=['accuracy'])
        return model

```

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[29]: history_smaller = buildModel(32, train_x).fit(train_x, train_y, epochs=50, ↪
↪batch_size=256, validation_data=(validation_x, validation_y))

```

```

Epoch 1/50
196/196          0s 976us/step -
accuracy: 0.8420 - loss: 0.6042 - val_accuracy: 0.9173 - val_loss: 0.2898
Epoch 2/50
196/196          0s 685us/step -
accuracy: 0.9233 - loss: 0.2656 - val_accuracy: 0.9319 - val_loss: 0.2330
Epoch 3/50
196/196          0s 679us/step -
accuracy: 0.9381 - loss: 0.2121 - val_accuracy: 0.9471 - val_loss: 0.1872
Epoch 4/50

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196/196 0s 702us/step -
 accuracy: 0.9475 - loss: 0.1825 - val_accuracy: 0.9516 - val_loss: 0.1685
 Epoch 5/50
 196/196 0s 707us/step -
 accuracy: 0.9533 - loss: 0.1628 - val_accuracy: 0.9558 - val_loss: 0.1580
 Epoch 6/50
 196/196 0s 734us/step -
 accuracy: 0.9573 - loss: 0.1470 - val_accuracy: 0.9563 - val_loss: 0.1537
 Epoch 7/50
 196/196 0s 918us/step -
 accuracy: 0.9610 - loss: 0.1348 - val_accuracy: 0.9530 - val_loss: 0.1626
 Epoch 8/50
 196/196 0s 717us/step -
 accuracy: 0.9638 - loss: 0.1248 - val_accuracy: 0.9587 - val_loss: 0.1409
 Epoch 9/50
 196/196 0s 708us/step -
 accuracy: 0.9660 - loss: 0.1164 - val_accuracy: 0.9596 - val_loss: 0.1436
 Epoch 10/50
 196/196 0s 670us/step -
 accuracy: 0.9681 - loss: 0.1089 - val_accuracy: 0.9565 - val_loss: 0.1435
 Epoch 11/50
 196/196 0s 678us/step -
 accuracy: 0.9699 - loss: 0.1037 - val_accuracy: 0.9602 - val_loss: 0.1326
 Epoch 12/50
 196/196 0s 667us/step -
 accuracy: 0.9710 - loss: 0.0977 - val_accuracy: 0.9589 - val_loss: 0.1402
 Epoch 13/50
 196/196 0s 735us/step -
 accuracy: 0.9734 - loss: 0.0925 - val_accuracy: 0.9588 - val_loss: 0.1350
 Epoch 14/50
 196/196 0s 892us/step -
 accuracy: 0.9743 - loss: 0.0882 - val_accuracy: 0.9628 - val_loss: 0.1312
 Epoch 15/50
 196/196 0s 828us/step -
 accuracy: 0.9750 - loss: 0.0841 - val_accuracy: 0.9631 - val_loss: 0.1301
 Epoch 16/50
 196/196 0s 746us/step -
 accuracy: 0.9766 - loss: 0.0803 - val_accuracy: 0.9623 - val_loss: 0.1360
 Epoch 17/50
 196/196 0s 671us/step -
 accuracy: 0.9772 - loss: 0.0762 - val_accuracy: 0.9641 - val_loss: 0.1262
 Epoch 18/50
 196/196 0s 677us/step -
 accuracy: 0.9789 - loss: 0.0735 - val_accuracy: 0.9624 - val_loss: 0.1309
 Epoch 19/50
 196/196 0s 639us/step -
 accuracy: 0.9792 - loss: 0.0701 - val_accuracy: 0.9639 - val_loss: 0.1272
 Epoch 20/50

196/196 0s 641us/step -
 accuracy: 0.9802 - loss: 0.0666 - val_accuracy: 0.9644 - val_loss: 0.1280
 Epoch 21/50
 196/196 0s 634us/step -
 accuracy: 0.9809 - loss: 0.0642 - val_accuracy: 0.9648 - val_loss: 0.1255
 Epoch 22/50
 196/196 0s 636us/step -
 accuracy: 0.9820 - loss: 0.0617 - val_accuracy: 0.9657 - val_loss: 0.1271
 Epoch 23/50
 196/196 0s 639us/step -
 accuracy: 0.9824 - loss: 0.0592 - val_accuracy: 0.9668 - val_loss: 0.1224
 Epoch 24/50
 196/196 0s 639us/step -
 accuracy: 0.9830 - loss: 0.0571 - val_accuracy: 0.9642 - val_loss: 0.1283
 Epoch 25/50
 196/196 0s 637us/step -
 accuracy: 0.9841 - loss: 0.0538 - val_accuracy: 0.9636 - val_loss: 0.1347
 Epoch 26/50
 196/196 0s 645us/step -
 accuracy: 0.9839 - loss: 0.0524 - val_accuracy: 0.9680 - val_loss: 0.1217
 Epoch 27/50
 196/196 0s 645us/step -
 accuracy: 0.9854 - loss: 0.0500 - val_accuracy: 0.9641 - val_loss: 0.1367
 Epoch 28/50
 196/196 0s 638us/step -
 accuracy: 0.9856 - loss: 0.0482 - val_accuracy: 0.9631 - val_loss: 0.1398
 Epoch 29/50
 196/196 0s 635us/step -
 accuracy: 0.9859 - loss: 0.0470 - val_accuracy: 0.9665 - val_loss: 0.1259
 Epoch 30/50
 196/196 0s 640us/step -
 accuracy: 0.9868 - loss: 0.0448 - val_accuracy: 0.9662 - val_loss: 0.1258
 Epoch 31/50
 196/196 0s 652us/step -
 accuracy: 0.9874 - loss: 0.0429 - val_accuracy: 0.9664 - val_loss: 0.1305
 Epoch 32/50
 196/196 0s 660us/step -
 accuracy: 0.9883 - loss: 0.0409 - val_accuracy: 0.9661 - val_loss: 0.1363
 Epoch 33/50
 196/196 0s 647us/step -
 accuracy: 0.9882 - loss: 0.0399 - val_accuracy: 0.9679 - val_loss: 0.1286
 Epoch 34/50
 196/196 0s 644us/step -
 accuracy: 0.9888 - loss: 0.0386 - val_accuracy: 0.9626 - val_loss: 0.1502
 Epoch 35/50
 196/196 0s 764us/step -
 accuracy: 0.9889 - loss: 0.0371 - val_accuracy: 0.9644 - val_loss: 0.1405
 Epoch 36/50

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196/196          0s 641us/step -
accuracy: 0.9899 - loss: 0.0351 - val_accuracy: 0.9641 - val_loss: 0.1431
Epoch 37/50
196/196          0s 639us/step -
accuracy: 0.9898 - loss: 0.0347 - val_accuracy: 0.9656 - val_loss: 0.1410
Epoch 38/50
196/196          0s 642us/step -
accuracy: 0.9905 - loss: 0.0335 - val_accuracy: 0.9594 - val_loss: 0.1628
Epoch 39/50
196/196          0s 635us/step -
accuracy: 0.9905 - loss: 0.0317 - val_accuracy: 0.9661 - val_loss: 0.1393
Epoch 40/50
196/196          0s 632us/step -
accuracy: 0.9909 - loss: 0.0307 - val_accuracy: 0.9678 - val_loss: 0.1381
Epoch 41/50
196/196          0s 629us/step -
accuracy: 0.9915 - loss: 0.0299 - val_accuracy: 0.9674 - val_loss: 0.1360
Epoch 42/50
196/196          0s 624us/step -
accuracy: 0.9918 - loss: 0.0285 - val_accuracy: 0.9632 - val_loss: 0.1540
Epoch 43/50
196/196          0s 625us/step -
accuracy: 0.9922 - loss: 0.0277 - val_accuracy: 0.9668 - val_loss: 0.1431
Epoch 44/50
196/196          0s 625us/step -
accuracy: 0.9929 - loss: 0.0263 - val_accuracy: 0.9645 - val_loss: 0.1484
Epoch 45/50
196/196          0s 632us/step -
accuracy: 0.9924 - loss: 0.0255 - val_accuracy: 0.9675 - val_loss: 0.1509
Epoch 46/50
196/196          0s 647us/step -
accuracy: 0.9932 - loss: 0.0244 - val_accuracy: 0.9599 - val_loss: 0.1714
Epoch 47/50
196/196          0s 625us/step -
accuracy: 0.9933 - loss: 0.0235 - val_accuracy: 0.9646 - val_loss: 0.1538
Epoch 48/50
196/196          0s 634us/step -
accuracy: 0.9934 - loss: 0.0227 - val_accuracy: 0.9653 - val_loss: 0.1528
Epoch 49/50
196/196          0s 639us/step -
accuracy: 0.9935 - loss: 0.0222 - val_accuracy: 0.9668 - val_loss: 0.1531
Epoch 50/50
196/196          0s 642us/step -
accuracy: 0.9939 - loss: 0.0211 - val_accuracy: 0.9684 - val_loss: 0.1510

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[30]: history_original = buildModel(64, train_x).fit(train_x, train_y, epochs=50,
↳ batch_size=256, validation_data=(validation_x, validation_y))

```

Epoch 1/50
196/196 0s 1ms/step -
accuracy: 0.8653 - loss: 0.4898 - val_accuracy: 0.9261 - val_loss: 0.2629

Epoch 2/50
196/196 0s 985us/step -
accuracy: 0.9348 - loss: 0.2225 - val_accuracy: 0.9338 - val_loss: 0.2310

Epoch 3/50
196/196 0s 891us/step -
accuracy: 0.9503 - loss: 0.1715 - val_accuracy: 0.9521 - val_loss: 0.1686

Epoch 4/50
196/196 0s 968us/step -
accuracy: 0.9580 - loss: 0.1414 - val_accuracy: 0.9633 - val_loss: 0.1339

Epoch 5/50
196/196 0s 875us/step -
accuracy: 0.9646 - loss: 0.1203 - val_accuracy: 0.9608 - val_loss: 0.1395

Epoch 6/50
196/196 0s 868us/step -
accuracy: 0.9689 - loss: 0.1046 - val_accuracy: 0.9638 - val_loss: 0.1302

Epoch 7/50
196/196 0s 831us/step -
accuracy: 0.9724 - loss: 0.0918 - val_accuracy: 0.9683 - val_loss: 0.1123

Epoch 8/50
196/196 0s 826us/step -
accuracy: 0.9755 - loss: 0.0809 - val_accuracy: 0.9673 - val_loss: 0.1120

Epoch 9/50
196/196 0s 836us/step -
accuracy: 0.9783 - loss: 0.0721 - val_accuracy: 0.9665 - val_loss: 0.1153

Epoch 10/50
196/196 0s 908us/step -
accuracy: 0.9799 - loss: 0.0650 - val_accuracy: 0.9686 - val_loss: 0.1048

Epoch 11/50
196/196 0s 914us/step -
accuracy: 0.9828 - loss: 0.0580 - val_accuracy: 0.9699 - val_loss: 0.1086

Epoch 12/50
196/196 0s 867us/step -
accuracy: 0.9841 - loss: 0.0525 - val_accuracy: 0.9697 - val_loss: 0.1037

Epoch 13/50
196/196 0s 826us/step -
accuracy: 0.9845 - loss: 0.0483 - val_accuracy: 0.9583 - val_loss: 0.1435

Epoch 14/50
196/196 0s 815us/step -
accuracy: 0.9876 - loss: 0.0426 - val_accuracy: 0.9716 - val_loss: 0.1011

Epoch 15/50
196/196 0s 825us/step -
accuracy: 0.9879 - loss: 0.0396 - val_accuracy: 0.9725 - val_loss: 0.0975

Epoch 16/50
196/196 0s 832us/step -
accuracy: 0.9894 - loss: 0.0355 - val_accuracy: 0.9718 - val_loss: 0.1057

Epoch 17/50
196/196 0s 813us/step -
accuracy: 0.9908 - loss: 0.0323 - val_accuracy: 0.9736 - val_loss: 0.1016
Epoch 18/50
196/196 0s 820us/step -
accuracy: 0.9914 - loss: 0.0295 - val_accuracy: 0.9704 - val_loss: 0.1092
Epoch 19/50
196/196 0s 850us/step -
accuracy: 0.9922 - loss: 0.0268 - val_accuracy: 0.9743 - val_loss: 0.1036
Epoch 20/50
196/196 0s 830us/step -
accuracy: 0.9931 - loss: 0.0242 - val_accuracy: 0.9746 - val_loss: 0.1028
Epoch 21/50
196/196 0s 834us/step -
accuracy: 0.9934 - loss: 0.0226 - val_accuracy: 0.9743 - val_loss: 0.1050
Epoch 22/50
196/196 0s 815us/step -
accuracy: 0.9949 - loss: 0.0200 - val_accuracy: 0.9738 - val_loss: 0.1078
Epoch 23/50
196/196 0s 854us/step -
accuracy: 0.9949 - loss: 0.0183 - val_accuracy: 0.9747 - val_loss: 0.1054
Epoch 24/50
196/196 0s 849us/step -
accuracy: 0.9954 - loss: 0.0164 - val_accuracy: 0.9749 - val_loss: 0.1086
Epoch 25/50
196/196 0s 821us/step -
accuracy: 0.9960 - loss: 0.0152 - val_accuracy: 0.9695 - val_loss: 0.1335
Epoch 26/50
196/196 0s 813us/step -
accuracy: 0.9964 - loss: 0.0135 - val_accuracy: 0.9727 - val_loss: 0.1254
Epoch 27/50
196/196 0s 834us/step -
accuracy: 0.9970 - loss: 0.0122 - val_accuracy: 0.9740 - val_loss: 0.1202
Epoch 28/50
196/196 0s 849us/step -
accuracy: 0.9971 - loss: 0.0111 - val_accuracy: 0.9738 - val_loss: 0.1238
Epoch 29/50
196/196 0s 812us/step -
accuracy: 0.9974 - loss: 0.0104 - val_accuracy: 0.9739 - val_loss: 0.1241
Epoch 30/50
196/196 0s 827us/step -
accuracy: 0.9974 - loss: 0.0097 - val_accuracy: 0.9742 - val_loss: 0.1212
Epoch 31/50
196/196 0s 803us/step -
accuracy: 0.9978 - loss: 0.0087 - val_accuracy: 0.9728 - val_loss: 0.1319
Epoch 32/50
196/196 0s 800us/step -
accuracy: 0.9979 - loss: 0.0082 - val_accuracy: 0.9688 - val_loss: 0.1533

Epoch 33/50
196/196 0s 875us/step -
accuracy: 0.9980 - loss: 0.0072 - val_accuracy: 0.9732 - val_loss: 0.1360

Epoch 34/50
196/196 0s 837us/step -
accuracy: 0.9984 - loss: 0.0064 - val_accuracy: 0.9744 - val_loss: 0.1345

Epoch 35/50
196/196 0s 857us/step -
accuracy: 0.9986 - loss: 0.0057 - val_accuracy: 0.9640 - val_loss: 0.1828

Epoch 36/50
196/196 0s 853us/step -
accuracy: 0.9984 - loss: 0.0059 - val_accuracy: 0.9722 - val_loss: 0.1410

Epoch 37/50
196/196 0s 841us/step -
accuracy: 0.9988 - loss: 0.0052 - val_accuracy: 0.9753 - val_loss: 0.1361

Epoch 38/50
196/196 0s 844us/step -
accuracy: 0.9991 - loss: 0.0041 - val_accuracy: 0.9736 - val_loss: 0.1380

Epoch 39/50
196/196 0s 854us/step -
accuracy: 0.9989 - loss: 0.0042 - val_accuracy: 0.9726 - val_loss: 0.1549

Epoch 40/50
196/196 0s 879us/step -
accuracy: 0.9987 - loss: 0.0044 - val_accuracy: 0.9725 - val_loss: 0.1541

Epoch 41/50
196/196 0s 847us/step -
accuracy: 0.9991 - loss: 0.0034 - val_accuracy: 0.9748 - val_loss: 0.1494

Epoch 42/50
196/196 0s 848us/step -
accuracy: 0.9992 - loss: 0.0035 - val_accuracy: 0.9753 - val_loss: 0.1609

Epoch 43/50
196/196 0s 842us/step -
accuracy: 0.9993 - loss: 0.0032 - val_accuracy: 0.9745 - val_loss: 0.1483

Epoch 44/50
196/196 0s 830us/step -
accuracy: 0.9994 - loss: 0.0026 - val_accuracy: 0.9737 - val_loss: 0.1602

Epoch 45/50
196/196 0s 830us/step -
accuracy: 0.9994 - loss: 0.0027 - val_accuracy: 0.9715 - val_loss: 0.1786

Epoch 46/50
196/196 0s 830us/step -
accuracy: 0.9993 - loss: 0.0026 - val_accuracy: 0.9737 - val_loss: 0.1623

Epoch 47/50
196/196 0s 829us/step -
accuracy: 0.9997 - loss: 0.0019 - val_accuracy: 0.9746 - val_loss: 0.1685

Epoch 48/50
196/196 0s 817us/step -
accuracy: 0.9998 - loss: 0.0015 - val_accuracy: 0.9740 - val_loss: 0.1719


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Epoch 49/50
196/196          0s 822us/step -
accuracy: 0.9996 - loss: 0.0018 - val_accuracy: 0.9744 - val_loss: 0.1656
Epoch 50/50
196/196          0s 848us/step -
accuracy: 0.9998 - loss: 0.0015 - val_accuracy: 0.9742 - val_loss: 0.1747
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```
[31]: history_larger = buildModel(512, train_x).fit(train_x, train_y, epochs=50,
↳ batch_size=256, validation_data=(validation_x, validation_y))
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```
Epoch 1/50
196/196          1s 3ms/step -
accuracy: 0.9014 - loss: 0.3233 - val_accuracy: 0.9566 - val_loss: 0.1475
Epoch 2/50
196/196          1s 3ms/step -
accuracy: 0.9641 - loss: 0.1141 - val_accuracy: 0.9590 - val_loss: 0.1302
Epoch 3/50
196/196          1s 3ms/step -
accuracy: 0.9773 - loss: 0.0713 - val_accuracy: 0.9695 - val_loss: 0.1030
Epoch 4/50
196/196          1s 3ms/step -
accuracy: 0.9848 - loss: 0.0497 - val_accuracy: 0.9751 - val_loss: 0.0876
Epoch 5/50
196/196          1s 3ms/step -
accuracy: 0.9889 - loss: 0.0352 - val_accuracy: 0.9731 - val_loss: 0.0935
Epoch 6/50
196/196          1s 3ms/step -
accuracy: 0.9920 - loss: 0.0257 - val_accuracy: 0.9756 - val_loss: 0.0884
Epoch 7/50
196/196          1s 3ms/step -
accuracy: 0.9943 - loss: 0.0179 - val_accuracy: 0.9770 - val_loss: 0.0886
Epoch 8/50
196/196          1s 3ms/step -
accuracy: 0.9960 - loss: 0.0129 - val_accuracy: 0.9802 - val_loss: 0.0807
Epoch 9/50
196/196          1s 3ms/step -
accuracy: 0.9973 - loss: 0.0095 - val_accuracy: 0.9794 - val_loss: 0.0818
Epoch 10/50
196/196          1s 3ms/step -
accuracy: 0.9979 - loss: 0.0068 - val_accuracy: 0.9758 - val_loss: 0.1113
Epoch 11/50
196/196          1s 3ms/step -
accuracy: 0.9984 - loss: 0.0051 - val_accuracy: 0.9805 - val_loss: 0.0913
Epoch 12/50
196/196          1s 3ms/step -
accuracy: 0.9990 - loss: 0.0040 - val_accuracy: 0.9796 - val_loss: 0.0947
Epoch 13/50
196/196          1s 3ms/step -
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accuracy: 0.9992 - loss: 0.0024 - val_accuracy: 0.9823 - val_loss: 0.0853
 Epoch 14/50
 196/196 1s 3ms/step -
 accuracy: 0.9997 - loss: 0.0015 - val_accuracy: 0.9826 - val_loss: 0.0837
 Epoch 15/50
 196/196 1s 3ms/step -
 accuracy: 0.9999 - loss: 7.0417e-04 - val_accuracy: 0.9823 - val_loss: 0.0868
 Epoch 16/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 1.9448e-04 - val_accuracy: 0.9836 - val_loss: 0.0845
 Epoch 17/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 1.1374e-04 - val_accuracy: 0.9841 - val_loss: 0.0859
 Epoch 18/50
 196/196 1s 3ms/step -
 accuracy: 0.9999 - loss: 2.9039e-04 - val_accuracy: 0.9838 - val_loss: 0.0884
 Epoch 19/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 1.3390e-04 - val_accuracy: 0.9840 - val_loss: 0.0877
 Epoch 20/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 6.2163e-05 - val_accuracy: 0.9838 - val_loss: 0.0898
 Epoch 21/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 5.3520e-05 - val_accuracy: 0.9837 - val_loss: 0.0899
 Epoch 22/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 4.8267e-05 - val_accuracy: 0.9842 - val_loss: 0.0903
 Epoch 23/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 4.3901e-05 - val_accuracy: 0.9838 - val_loss: 0.0910
 Epoch 24/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 4.0845e-05 - val_accuracy: 0.9835 - val_loss: 0.0916
 Epoch 25/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 3.8212e-05 - val_accuracy: 0.9843 - val_loss: 0.0921
 Epoch 26/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 3.5819e-05 - val_accuracy: 0.9838 - val_loss: 0.0922
 Epoch 27/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 3.3889e-05 - val_accuracy: 0.9839 - val_loss: 0.0930
 Epoch 28/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 3.2098e-05 - val_accuracy: 0.9840 - val_loss: 0.0935
 Epoch 29/50
 196/196 1s 3ms/step -

accuracy: 1.0000 - loss: 3.0715e-05 - val_accuracy: 0.9839 - val_loss: 0.0935
 Epoch 30/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 2.9225e-05 - val_accuracy: 0.9842 - val_loss: 0.0939
 Epoch 31/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 2.7982e-05 - val_accuracy: 0.9840 - val_loss: 0.0943
 Epoch 32/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 2.6915e-05 - val_accuracy: 0.9840 - val_loss: 0.0946
 Epoch 33/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 2.5832e-05 - val_accuracy: 0.9842 - val_loss: 0.0949
 Epoch 34/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 2.4865e-05 - val_accuracy: 0.9840 - val_loss: 0.0953
 Epoch 35/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 2.3979e-05 - val_accuracy: 0.9839 - val_loss: 0.0957
 Epoch 36/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 2.3156e-05 - val_accuracy: 0.9840 - val_loss: 0.0959
 Epoch 37/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 2.2438e-05 - val_accuracy: 0.9840 - val_loss: 0.0963
 Epoch 38/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 2.1677e-05 - val_accuracy: 0.9841 - val_loss: 0.0964
 Epoch 39/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 2.1083e-05 - val_accuracy: 0.9840 - val_loss: 0.0967
 Epoch 40/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 2.0449e-05 - val_accuracy: 0.9841 - val_loss: 0.0970
 Epoch 41/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 1.9844e-05 - val_accuracy: 0.9841 - val_loss: 0.0970
 Epoch 42/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 1.9321e-05 - val_accuracy: 0.9842 - val_loss: 0.0973
 Epoch 43/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 1.8766e-05 - val_accuracy: 0.9842 - val_loss: 0.0975
 Epoch 44/50
 196/196 1s 3ms/step -
 accuracy: 1.0000 - loss: 1.8282e-05 - val_accuracy: 0.9844 - val_loss: 0.0978
 Epoch 45/50
 196/196 1s 3ms/step -

```

accuracy: 1.0000 - loss: 1.7818e-05 - val_accuracy: 0.9842 - val_loss: 0.0980
Epoch 46/50
196/196          1s 3ms/step -
accuracy: 1.0000 - loss: 1.7397e-05 - val_accuracy: 0.9841 - val_loss: 0.0982
Epoch 47/50
196/196          1s 3ms/step -
accuracy: 1.0000 - loss: 1.6960e-05 - val_accuracy: 0.9842 - val_loss: 0.0983
Epoch 48/50
196/196          1s 3ms/step -
accuracy: 1.0000 - loss: 1.6544e-05 - val_accuracy: 0.9840 - val_loss: 0.0987
Epoch 49/50
196/196          1s 3ms/step -
accuracy: 1.0000 - loss: 1.6201e-05 - val_accuracy: 0.9840 - val_loss: 0.0988
Epoch 50/50
196/196          1s 3ms/step -
accuracy: 1.0000 - loss: 1.5827e-05 - val_accuracy: 0.9841 - val_loss: 0.0990

```

```

[32]: def plotLoss(history,name) :
        loss = history.history['loss']
        val_loss = history.history['val_loss']
        epochs = range(1, len(loss) + 1)
        plt.plot(epochs,loss,'r',label='Training loss')
        plt.plot(epochs,val_loss,'b',label='Validation loss')
        plt.title('Training and Validation Loss ' + name)
        plt.xlabel('Epochs')
        plt.ylabel('Loss')
        plt.legend()
        plt.show()

```

```

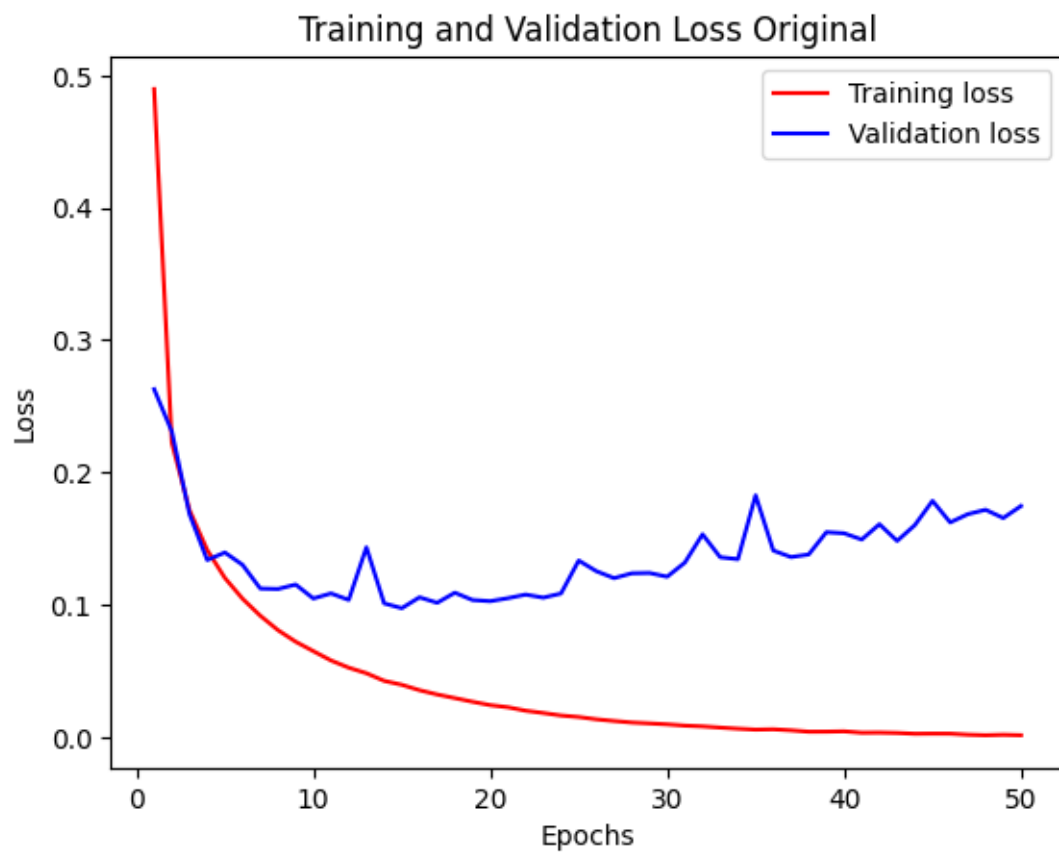
[33]: def plotAcc(history,name) :
        accuracy = history.history['accuracy']
        val_accuracy = history.history['val_accuracy']
        epochs = range(1 , len(accuracy) + 1)
        plt.plot(epochs,accuracy,'r',label='Training Accuracy')
        plt.plot(epochs,val_accuracy,'b',label='Validation Accuracy')
        plt.title('Training and Validation Accuracy ' + name)
        plt.xlabel('Epochs')
        plt.ylabel('Accuracy')
        plt.legend()
        plt.show()

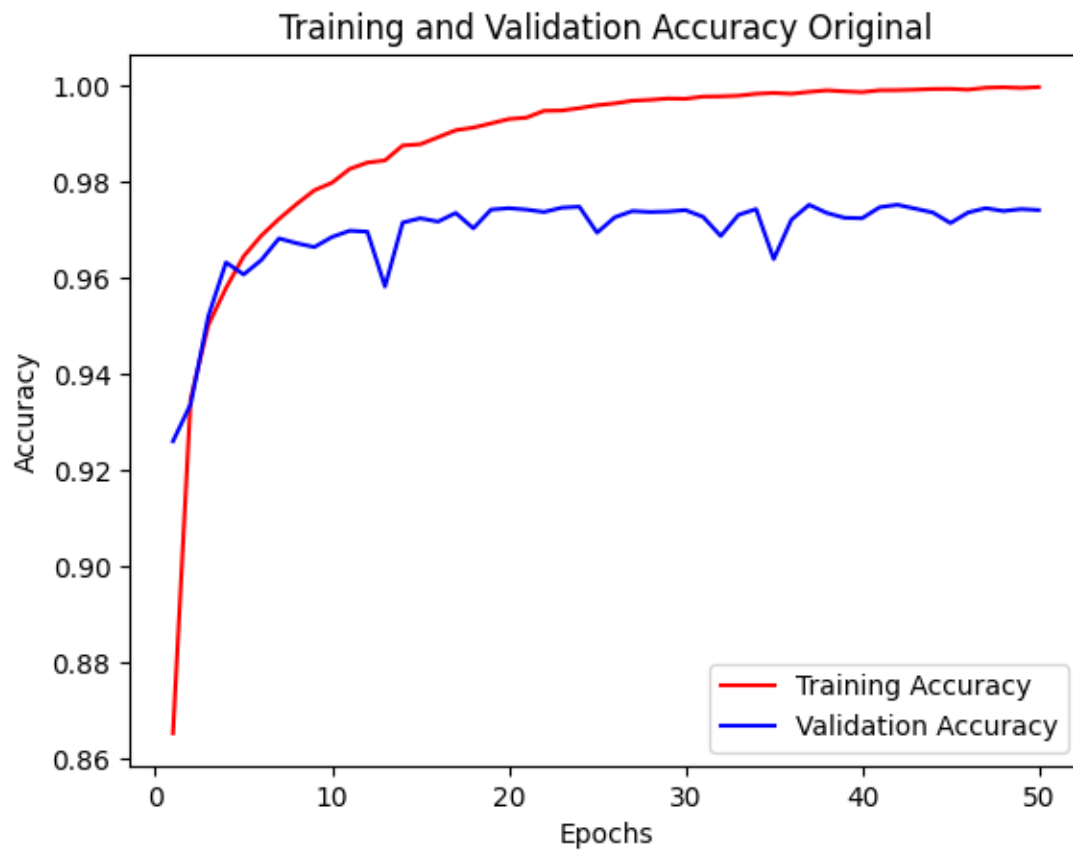
```

```

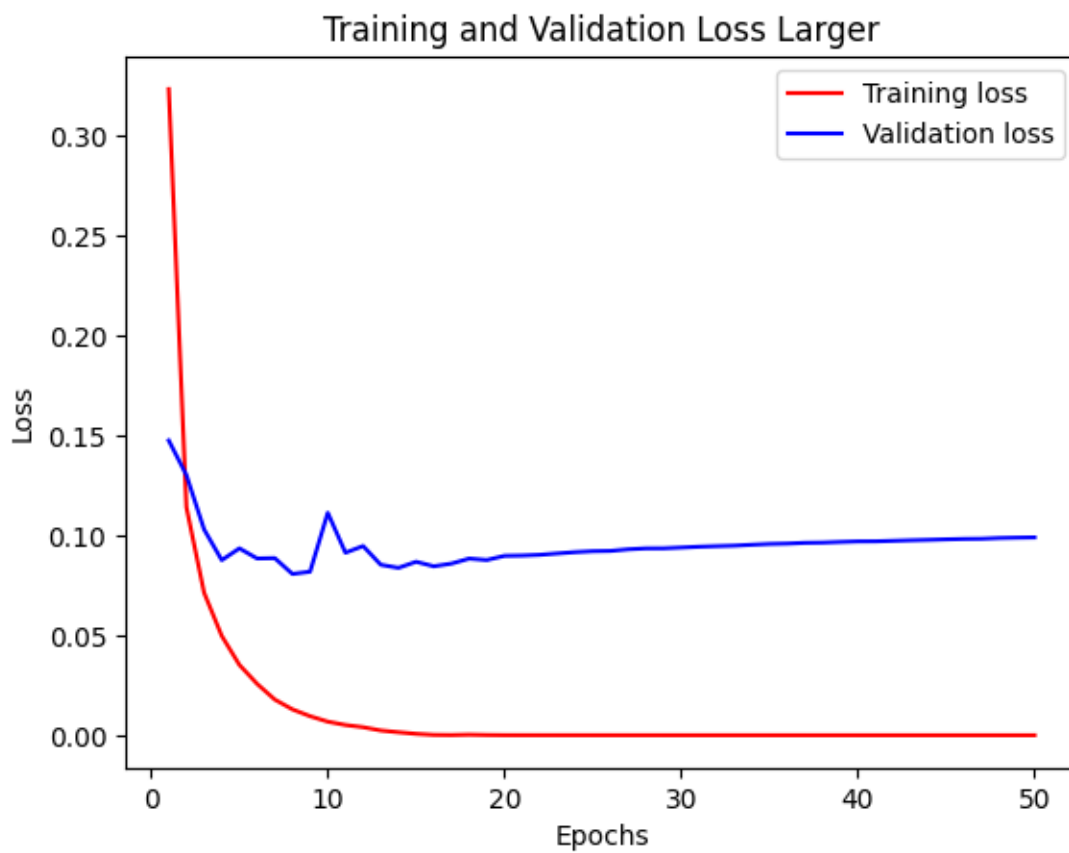
[34]: plotLoss(history_original,"Original")
        plotAcc(history_original,"Original")

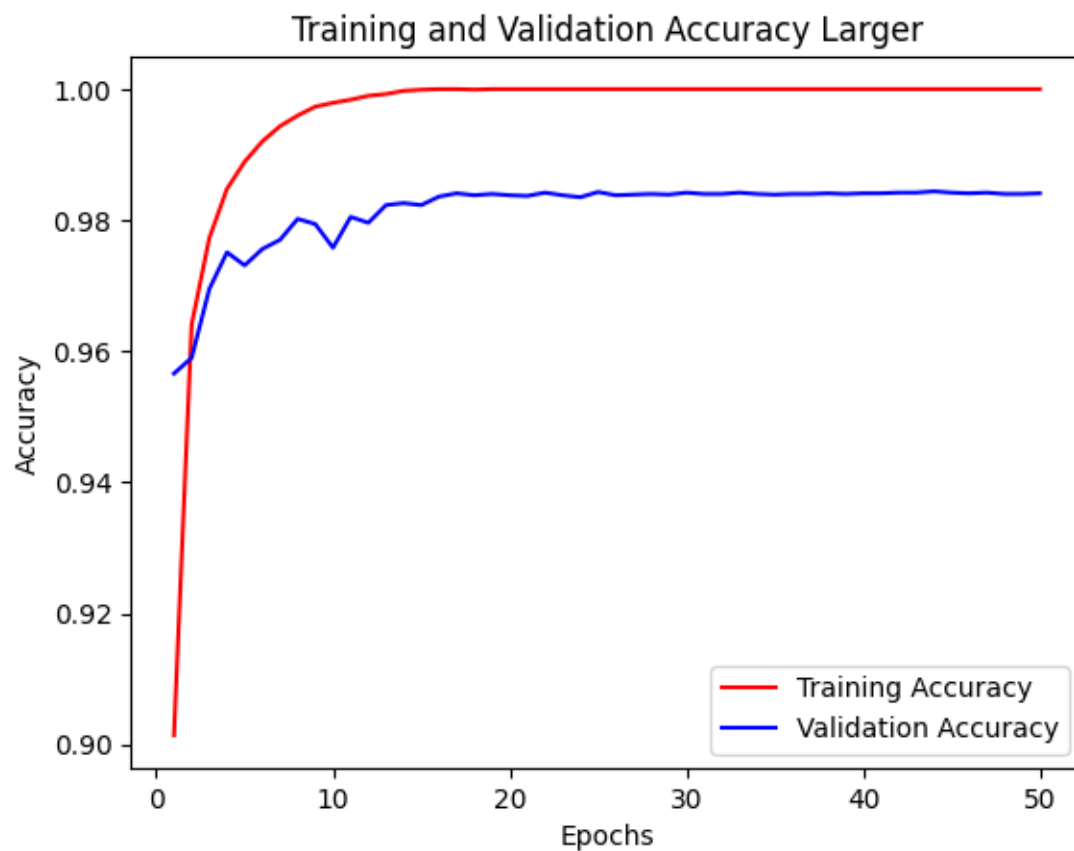
```



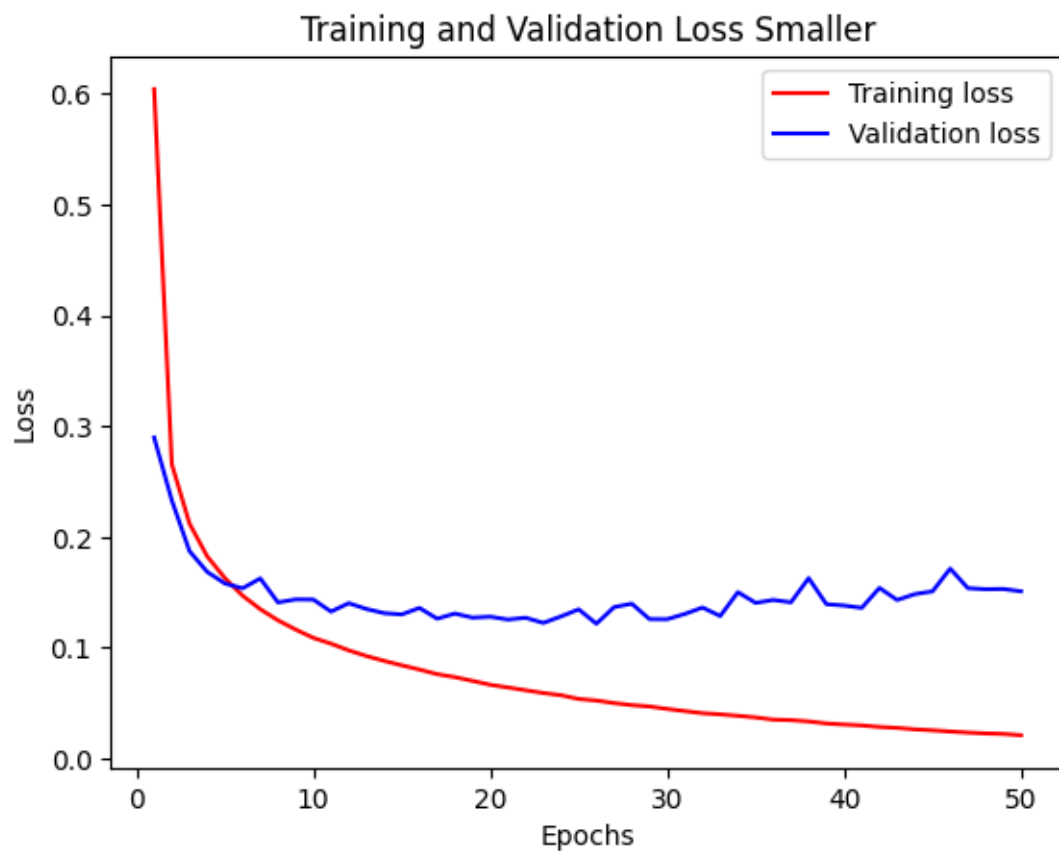


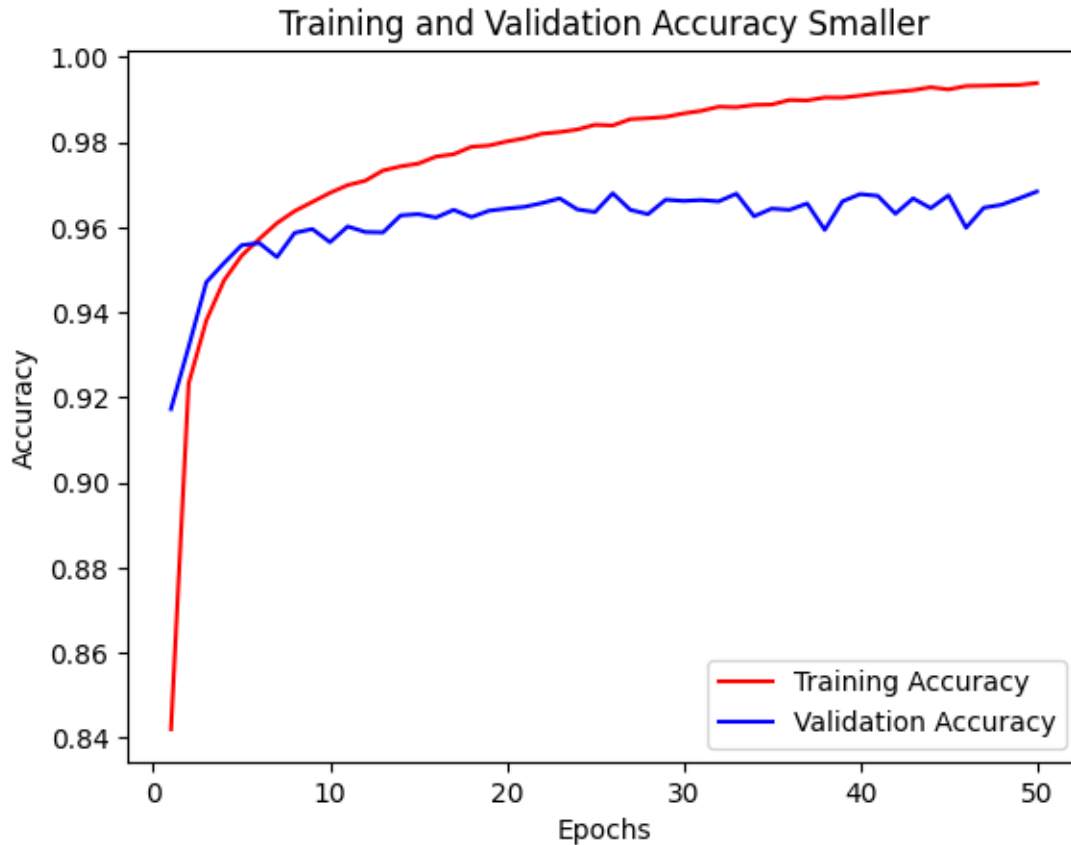
```
[35]: plotLoss(history_larger, "Larger")  
      plotAcc(history_larger, "Larger")
```





```
[36]: plotLoss(history_smaller,"Smaller")  
      plotAcc(history_smaller,"Smaller")
```



```
[37]: history_Dropout = buildModelDropout(64 , train_x).
      ↪fit(train_x,train_y,epochs=50,batch_size=256,validation_data=(validation_x,validation_y))
```

```
Epoch 1/50
196/196          1s 1ms/step -
accuracy: 0.6272 - loss: 1.1360 - val_accuracy: 0.9072 - val_loss: 0.3389
Epoch 2/50
196/196          0s 963us/step -
accuracy: 0.8255 - loss: 0.5896 - val_accuracy: 0.9299 - val_loss: 0.2502
Epoch 3/50
196/196          0s 944us/step -
accuracy: 0.8605 - loss: 0.4835 - val_accuracy: 0.9344 - val_loss: 0.2305
Epoch 4/50
196/196          0s 935us/step -
accuracy: 0.8769 - loss: 0.4373 - val_accuracy: 0.9403 - val_loss: 0.2053
Epoch 5/50
196/196          0s 968us/step -
accuracy: 0.8872 - loss: 0.3985 - val_accuracy: 0.9457 - val_loss: 0.1917
Epoch 6/50
196/196          0s 963us/step -
```

accuracy: 0.8940 - loss: 0.3769 - val_accuracy: 0.9485 - val_loss: 0.1878
 Epoch 7/50
 196/196 0s 987us/step -
 accuracy: 0.8985 - loss: 0.3561 - val_accuracy: 0.9524 - val_loss: 0.1743
 Epoch 8/50
 196/196 0s 954us/step -
 accuracy: 0.9014 - loss: 0.3461 - val_accuracy: 0.9560 - val_loss: 0.1648
 Epoch 9/50
 196/196 0s 1ms/step -
 accuracy: 0.9063 - loss: 0.3320 - val_accuracy: 0.9559 - val_loss: 0.1650
 Epoch 10/50
 196/196 0s 1ms/step -
 accuracy: 0.9088 - loss: 0.3225 - val_accuracy: 0.9579 - val_loss: 0.1556
 Epoch 11/50
 196/196 0s 978us/step -
 accuracy: 0.9098 - loss: 0.3197 - val_accuracy: 0.9592 - val_loss: 0.1529
 Epoch 12/50
 196/196 0s 976us/step -
 accuracy: 0.9127 - loss: 0.3089 - val_accuracy: 0.9608 - val_loss: 0.1551
 Epoch 13/50
 196/196 0s 954us/step -
 accuracy: 0.9145 - loss: 0.3049 - val_accuracy: 0.9621 - val_loss: 0.1475
 Epoch 14/50
 196/196 0s 973us/step -
 accuracy: 0.9156 - loss: 0.2943 - val_accuracy: 0.9619 - val_loss: 0.1471
 Epoch 15/50
 196/196 0s 989us/step -
 accuracy: 0.9182 - loss: 0.2933 - val_accuracy: 0.9629 - val_loss: 0.1431
 Epoch 16/50
 196/196 0s 938us/step -
 accuracy: 0.9175 - loss: 0.2896 - val_accuracy: 0.9625 - val_loss: 0.1458
 Epoch 17/50
 196/196 0s 962us/step -
 accuracy: 0.9212 - loss: 0.2826 - val_accuracy: 0.9643 - val_loss: 0.1415
 Epoch 18/50
 196/196 0s 967us/step -
 accuracy: 0.9233 - loss: 0.2748 - val_accuracy: 0.9651 - val_loss: 0.1397
 Epoch 19/50
 196/196 0s 948us/step -
 accuracy: 0.9214 - loss: 0.2796 - val_accuracy: 0.9675 - val_loss: 0.1353
 Epoch 20/50
 196/196 0s 1ms/step -
 accuracy: 0.9241 - loss: 0.2712 - val_accuracy: 0.9647 - val_loss: 0.1402
 Epoch 21/50
 196/196 0s 1ms/step -
 accuracy: 0.9239 - loss: 0.2737 - val_accuracy: 0.9663 - val_loss: 0.1354
 Epoch 22/50
 196/196 0s 1ms/step -

accuracy: 0.9238 - loss: 0.2676 - val_accuracy: 0.9665 - val_loss: 0.1378
 Epoch 23/50
 196/196 0s 1ms/step -
 accuracy: 0.9262 - loss: 0.2616 - val_accuracy: 0.9669 - val_loss: 0.1378
 Epoch 24/50
 196/196 0s 1ms/step -
 accuracy: 0.9252 - loss: 0.2644 - val_accuracy: 0.9648 - val_loss: 0.1365
 Epoch 25/50
 196/196 0s 1ms/step -
 accuracy: 0.9264 - loss: 0.2583 - val_accuracy: 0.9662 - val_loss: 0.1350
 Epoch 26/50
 196/196 0s 1ms/step -
 accuracy: 0.9281 - loss: 0.2564 - val_accuracy: 0.9668 - val_loss: 0.1311
 Epoch 27/50
 196/196 0s 1ms/step -
 accuracy: 0.9289 - loss: 0.2566 - val_accuracy: 0.9669 - val_loss: 0.1356
 Epoch 28/50
 196/196 0s 1ms/step -
 accuracy: 0.9289 - loss: 0.2500 - val_accuracy: 0.9668 - val_loss: 0.1344
 Epoch 29/50
 196/196 0s 1ms/step -
 accuracy: 0.9307 - loss: 0.2493 - val_accuracy: 0.9650 - val_loss: 0.1431
 Epoch 30/50
 196/196 0s 1ms/step -
 accuracy: 0.9288 - loss: 0.2487 - val_accuracy: 0.9682 - val_loss: 0.1382
 Epoch 31/50
 196/196 0s 1ms/step -
 accuracy: 0.9302 - loss: 0.2488 - val_accuracy: 0.9671 - val_loss: 0.1373
 Epoch 32/50
 196/196 0s 1ms/step -
 accuracy: 0.9281 - loss: 0.2564 - val_accuracy: 0.9665 - val_loss: 0.1382
 Epoch 33/50
 196/196 0s 999us/step -
 accuracy: 0.9301 - loss: 0.2496 - val_accuracy: 0.9678 - val_loss: 0.1334
 Epoch 34/50
 196/196 0s 950us/step -
 accuracy: 0.9299 - loss: 0.2468 - val_accuracy: 0.9673 - val_loss: 0.1364
 Epoch 35/50
 196/196 0s 1ms/step -
 accuracy: 0.9297 - loss: 0.2490 - val_accuracy: 0.9690 - val_loss: 0.1340
 Epoch 36/50
 196/196 0s 1ms/step -
 accuracy: 0.9311 - loss: 0.2429 - val_accuracy: 0.9683 - val_loss: 0.1353
 Epoch 37/50
 196/196 0s 994us/step -
 accuracy: 0.9325 - loss: 0.2370 - val_accuracy: 0.9674 - val_loss: 0.1369
 Epoch 38/50
 196/196 0s 934us/step -

```

accuracy: 0.9304 - loss: 0.2432 - val_accuracy: 0.9667 - val_loss: 0.1443
Epoch 39/50
196/196          0s 919us/step -
accuracy: 0.9322 - loss: 0.2392 - val_accuracy: 0.9675 - val_loss: 0.1423
Epoch 40/50
196/196          0s 1ms/step -
accuracy: 0.9309 - loss: 0.2411 - val_accuracy: 0.9688 - val_loss: 0.1358
Epoch 41/50
196/196          0s 1ms/step -
accuracy: 0.9336 - loss: 0.2339 - val_accuracy: 0.9665 - val_loss: 0.1441
Epoch 42/50
196/196          0s 1ms/step -
accuracy: 0.9324 - loss: 0.2355 - val_accuracy: 0.9688 - val_loss: 0.1361
Epoch 43/50
196/196          0s 955us/step -
accuracy: 0.9330 - loss: 0.2378 - val_accuracy: 0.9665 - val_loss: 0.1487
Epoch 44/50
196/196          0s 963us/step -
accuracy: 0.9334 - loss: 0.2355 - val_accuracy: 0.9680 - val_loss: 0.1392
Epoch 45/50
196/196          0s 1ms/step -
accuracy: 0.9335 - loss: 0.2346 - val_accuracy: 0.9679 - val_loss: 0.1396
Epoch 46/50
196/196          0s 956us/step -
accuracy: 0.9344 - loss: 0.2322 - val_accuracy: 0.9668 - val_loss: 0.1437
Epoch 47/50
196/196          0s 964us/step -
accuracy: 0.9329 - loss: 0.2291 - val_accuracy: 0.9678 - val_loss: 0.1398
Epoch 48/50
196/196          0s 935us/step -
accuracy: 0.9343 - loss: 0.2315 - val_accuracy: 0.9672 - val_loss: 0.1392
Epoch 49/50
196/196          0s 935us/step -
accuracy: 0.9332 - loss: 0.2348 - val_accuracy: 0.9685 - val_loss: 0.1489
Epoch 50/50
196/196          0s 940us/step -
accuracy: 0.9347 - loss: 0.2331 - val_accuracy: 0.9682 - val_loss: 0.1420

```

```

[38]: history-Regularized = buildModelRegularize(64 , train_x).
      ↪ fit(train_x,train_y,epochs=50,batch_size=256,validation_data=(validation_x,validation_y))

```

```

Epoch 1/50
196/196          1s 1ms/step -
accuracy: 0.8556 - loss: 0.6809 - val_accuracy: 0.9130 - val_loss: 0.4296
Epoch 2/50
196/196          0s 845us/step -
accuracy: 0.9261 - loss: 0.3885 - val_accuracy: 0.9292 - val_loss: 0.3608
Epoch 3/50

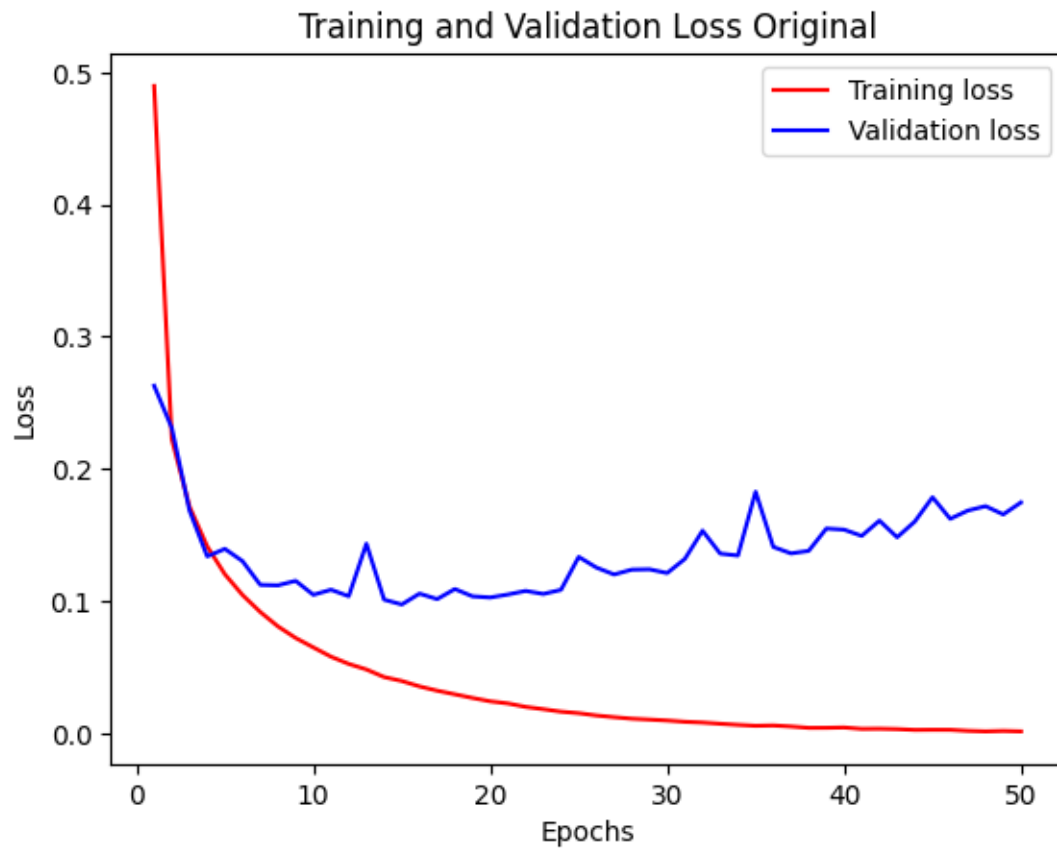
```

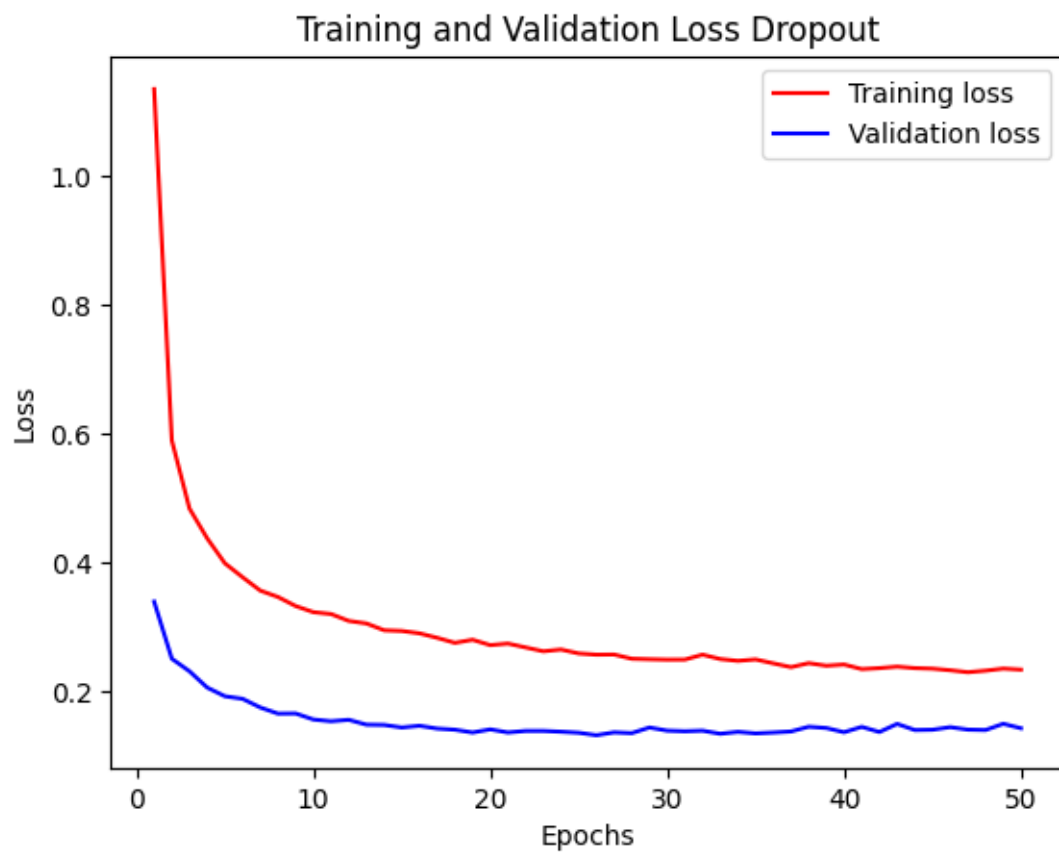
196/196 0s 824us/step -
 accuracy: 0.9399 - loss: 0.3295 - val_accuracy: 0.9312 - val_loss: 0.3500
 Epoch 4/50
 196/196 0s 810us/step -
 accuracy: 0.9487 - loss: 0.2910 - val_accuracy: 0.9491 - val_loss: 0.2846
 Epoch 5/50
 196/196 0s 828us/step -
 accuracy: 0.9552 - loss: 0.2627 - val_accuracy: 0.9550 - val_loss: 0.2582
 Epoch 6/50
 196/196 0s 832us/step -
 accuracy: 0.9600 - loss: 0.2418 - val_accuracy: 0.9610 - val_loss: 0.2337
 Epoch 7/50
 196/196 0s 834us/step -
 accuracy: 0.9634 - loss: 0.2269 - val_accuracy: 0.9641 - val_loss: 0.2270
 Epoch 8/50
 196/196 0s 865us/step -
 accuracy: 0.9669 - loss: 0.2135 - val_accuracy: 0.9680 - val_loss: 0.2090
 Epoch 9/50
 196/196 0s 840us/step -
 accuracy: 0.9690 - loss: 0.2019 - val_accuracy: 0.9550 - val_loss: 0.2406
 Epoch 10/50
 196/196 0s 843us/step -
 accuracy: 0.9710 - loss: 0.1929 - val_accuracy: 0.9609 - val_loss: 0.2201
 Epoch 11/50
 196/196 0s 850us/step -
 accuracy: 0.9717 - loss: 0.1852 - val_accuracy: 0.9679 - val_loss: 0.2006
 Epoch 12/50
 196/196 0s 845us/step -
 accuracy: 0.9740 - loss: 0.1771 - val_accuracy: 0.9693 - val_loss: 0.1865
 Epoch 13/50
 196/196 0s 848us/step -
 accuracy: 0.9749 - loss: 0.1715 - val_accuracy: 0.9600 - val_loss: 0.2208
 Epoch 14/50
 196/196 0s 863us/step -
 accuracy: 0.9760 - loss: 0.1664 - val_accuracy: 0.9690 - val_loss: 0.1862
 Epoch 15/50
 196/196 0s 866us/step -
 accuracy: 0.9764 - loss: 0.1614 - val_accuracy: 0.9668 - val_loss: 0.1921
 Epoch 16/50
 196/196 0s 847us/step -
 accuracy: 0.9785 - loss: 0.1560 - val_accuracy: 0.9611 - val_loss: 0.2085
 Epoch 17/50
 196/196 0s 842us/step -
 accuracy: 0.9783 - loss: 0.1528 - val_accuracy: 0.9726 - val_loss: 0.1763
 Epoch 18/50
 196/196 0s 848us/step -
 accuracy: 0.9795 - loss: 0.1489 - val_accuracy: 0.9692 - val_loss: 0.1798
 Epoch 19/50

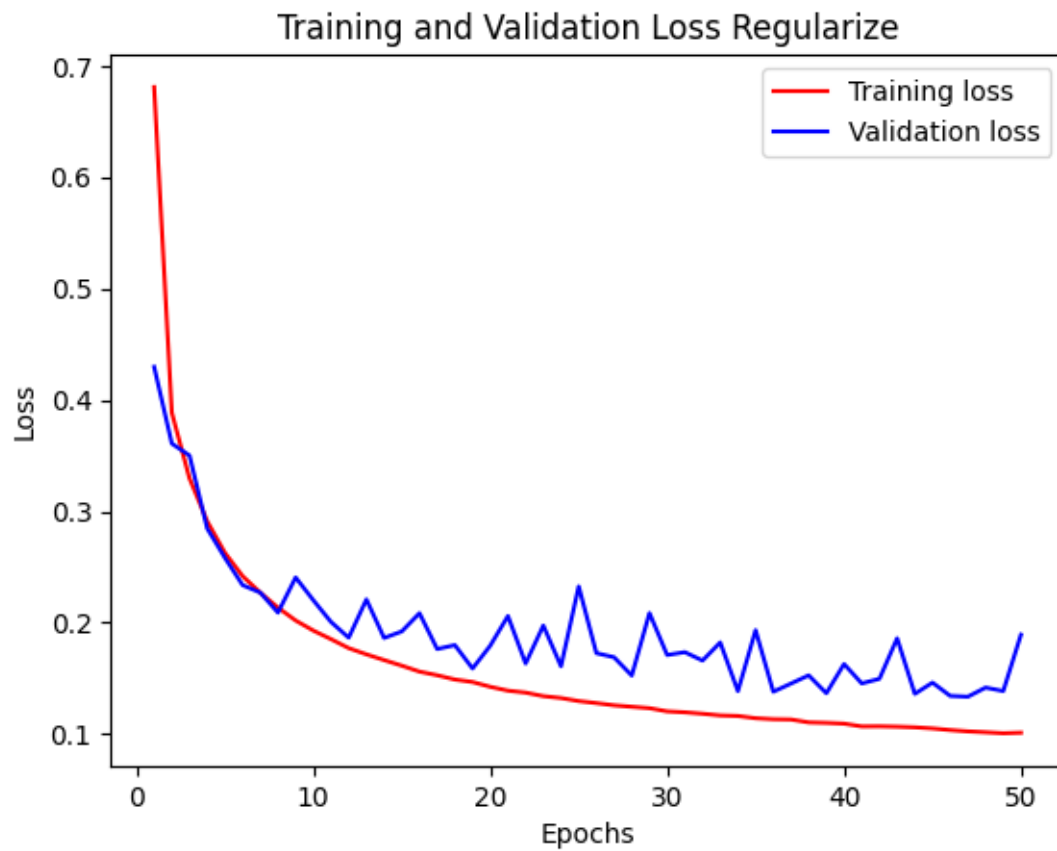
196/196 0s 839us/step -
 accuracy: 0.9796 - loss: 0.1467 - val_accuracy: 0.9758 - val_loss: 0.1589
 Epoch 20/50
 196/196 0s 843us/step -
 accuracy: 0.9801 - loss: 0.1424 - val_accuracy: 0.9694 - val_loss: 0.1793
 Epoch 21/50
 196/196 0s 846us/step -
 accuracy: 0.9805 - loss: 0.1389 - val_accuracy: 0.9583 - val_loss: 0.2060
 Epoch 22/50
 196/196 0s 860us/step -
 accuracy: 0.9810 - loss: 0.1372 - val_accuracy: 0.9732 - val_loss: 0.1634
 Epoch 23/50
 196/196 0s 837us/step -
 accuracy: 0.9817 - loss: 0.1340 - val_accuracy: 0.9603 - val_loss: 0.1974
 Epoch 24/50
 196/196 0s 841us/step -
 accuracy: 0.9819 - loss: 0.1323 - val_accuracy: 0.9736 - val_loss: 0.1608
 Epoch 25/50
 196/196 0s 844us/step -
 accuracy: 0.9818 - loss: 0.1296 - val_accuracy: 0.9493 - val_loss: 0.2326
 Epoch 26/50
 196/196 0s 844us/step -
 accuracy: 0.9830 - loss: 0.1278 - val_accuracy: 0.9686 - val_loss: 0.1726
 Epoch 27/50
 196/196 0s 829us/step -
 accuracy: 0.9829 - loss: 0.1258 - val_accuracy: 0.9687 - val_loss: 0.1689
 Epoch 28/50
 196/196 0s 837us/step -
 accuracy: 0.9832 - loss: 0.1245 - val_accuracy: 0.9740 - val_loss: 0.1525
 Epoch 29/50
 196/196 0s 838us/step -
 accuracy: 0.9837 - loss: 0.1232 - val_accuracy: 0.9549 - val_loss: 0.2086
 Epoch 30/50
 196/196 0s 827us/step -
 accuracy: 0.9834 - loss: 0.1203 - val_accuracy: 0.9686 - val_loss: 0.1709
 Epoch 31/50
 196/196 0s 839us/step -
 accuracy: 0.9843 - loss: 0.1195 - val_accuracy: 0.9688 - val_loss: 0.1735
 Epoch 32/50
 196/196 0s 808us/step -
 accuracy: 0.9840 - loss: 0.1182 - val_accuracy: 0.9687 - val_loss: 0.1660
 Epoch 33/50
 196/196 0s 809us/step -
 accuracy: 0.9842 - loss: 0.1167 - val_accuracy: 0.9659 - val_loss: 0.1822
 Epoch 34/50
 196/196 0s 816us/step -
 accuracy: 0.9838 - loss: 0.1162 - val_accuracy: 0.9772 - val_loss: 0.1385
 Epoch 35/50

196/196 0s 817us/step -
 accuracy: 0.9843 - loss: 0.1142 - val_accuracy: 0.9604 - val_loss: 0.1933
 Epoch 36/50
 196/196 0s 819us/step -
 accuracy: 0.9854 - loss: 0.1133 - val_accuracy: 0.9765 - val_loss: 0.1380
 Epoch 37/50
 196/196 0s 828us/step -
 accuracy: 0.9850 - loss: 0.1131 - val_accuracy: 0.9747 - val_loss: 0.1453
 Epoch 38/50
 196/196 0s 808us/step -
 accuracy: 0.9856 - loss: 0.1104 - val_accuracy: 0.9728 - val_loss: 0.1527
 Epoch 39/50
 196/196 0s 815us/step -
 accuracy: 0.9860 - loss: 0.1100 - val_accuracy: 0.9758 - val_loss: 0.1366
 Epoch 40/50
 196/196 0s 814us/step -
 accuracy: 0.9855 - loss: 0.1093 - val_accuracy: 0.9692 - val_loss: 0.1628
 Epoch 41/50
 196/196 0s 831us/step -
 accuracy: 0.9862 - loss: 0.1068 - val_accuracy: 0.9743 - val_loss: 0.1454
 Epoch 42/50
 196/196 0s 814us/step -
 accuracy: 0.9860 - loss: 0.1069 - val_accuracy: 0.9732 - val_loss: 0.1496
 Epoch 43/50
 196/196 0s 816us/step -
 accuracy: 0.9857 - loss: 0.1066 - val_accuracy: 0.9609 - val_loss: 0.1859
 Epoch 44/50
 196/196 0s 822us/step -
 accuracy: 0.9862 - loss: 0.1060 - val_accuracy: 0.9769 - val_loss: 0.1361
 Epoch 45/50
 196/196 0s 807us/step -
 accuracy: 0.9861 - loss: 0.1050 - val_accuracy: 0.9740 - val_loss: 0.1462
 Epoch 46/50
 196/196 0s 813us/step -
 accuracy: 0.9867 - loss: 0.1036 - val_accuracy: 0.9768 - val_loss: 0.1342
 Epoch 47/50
 196/196 0s 814us/step -
 accuracy: 0.9869 - loss: 0.1024 - val_accuracy: 0.9780 - val_loss: 0.1335
 Epoch 48/50
 196/196 0s 826us/step -
 accuracy: 0.9865 - loss: 0.1016 - val_accuracy: 0.9752 - val_loss: 0.1417
 Epoch 49/50
 196/196 0s 937us/step -
 accuracy: 0.9869 - loss: 0.1007 - val_accuracy: 0.9754 - val_loss: 0.1386
 Epoch 50/50
 196/196 0s 825us/step -
 accuracy: 0.9869 - loss: 0.1012 - val_accuracy: 0.9608 - val_loss: 0.1892


```
[39]: plotLoss(history_original, 'Original')  
      plotLoss(history_Dropout, 'Dropout')  
      plotLoss(history-Regularized, 'Regularize')
```







```
[40]: plotAcc(history_original, 'Original')  
      plotAcc(history_Dropout, 'Dropout')  
      plotAcc(history_Regularized, 'Regularize')
```

